How to use: smFISH_advanced_quantifications.py

Distance of dendritic mRNAs from the soma, localization indices, mRNAs per spot and neuron morphology, each with both statistical analyses. pipeline version 4 (October 2026).

> **Needs:** Fish_spots_batch_v4.py and smFISH_statistics_superplots.py in the same folder, and a finished Fish_spots_batch_v4 run with your final settings (it uses the same masks and the filtered spots).

1. Running it

```
python smFISH_advanced_quantifications.py
-> select the top folder ("biological replicates")
```

First run 30-60 min (distance maps and skeletons, parallel, cached); reruns minutes. Results: `_density_results/advanced_quantifications/`.

> If you change dilation or PROB_THRESHOLD, rerun Fish_spots_batch_v4.py first so both use the same masks.

2. Metrics

Metric	Definition	Read as
Distance profile	spot density in 5 µm bins of path distance from the soma through the dendrite mask	absolute and relative (1 = mean of that dendrite)
Distal index	mean spot distance / mean distance of all dendrite pixels	>1 distal, <1 proximal
Median spot distance	median path distance of dendritic spots (µm)	
Dendrite/soma, dendrite/axon ratio	density ratio per image	compares probes independent of labelling efficiency
mRNAs per spot	integrated intensity / single-mRNA unit of the pair	≈1 single mRNA; ≥1.5 multi-mRNA spot
Fraction multi-mRNA spots	spots ≥ MULTI_COPY_THRESHOLD units	granule content
mRNA copies per neuron	sum of units in soma + dendrites	
Soma area, dendrite length, primary dendrites, reach	tree connected to the soma; pruned skeleton	quality check: should not differ control vs protease

3. Settings

Setting	Default	Meaning
BIN_UM / MAX_DISTANCE_UM	5 / 150	profile bin width and plotted range
MIN_BIN_AREA_UM2	5	bins with less dendrite area in an image are ignored
MULTI_COPY_THRESHOLD	1.5	multi-mRNA threshold (units)
MIN_SPOTS_FOR_UNIT	30	fewer neurite spots → unit from all compartments
SPUR_UM / HOLE_UM2	3 / 2	skeleton cleaning (spurs removed, small holes filled)

4. Outputs

- `advanced_quantifications.xlsx`: per image, replicate means, DIV and control/protease statistics (t-test and robust model), distance profiles per image, single-mRNA unit per pair, methods.
- `PROFILE_distance_from_soma_<condition>(_relative).pdf`: profiles per probe, DIV9 vs DIV27.
- `robust_linear_model/` and `t_tests/`: one SuperPlot per metric.

5. Caveats

- mRNAs per spot assumes most counted spots are single mRNAs; check `single_mRNA_unit_per_pair` for implausible values (pairs with residual overdetection).
- Soma spots merge in the projection; somatic mRNAs per spot are overestimated.
- Branch points are listed per image only; on projected masks they are not reliable enough for statistics.